



CODEx / REUSABLE BUILD PROMPT

Premium scroll-controlled video component

A production-ready brief for building cinematic, responsive, high-resolution video sequences controlled by natural page scroll.

SMOOTH BIDIRECTIONAL SCRUBBING

ADAPTIVE 8K / 4K / MOBILE MEDIA

ACCESSIBLE + PERFORMANCE-AWARE

Version 1.0 / Ready to copy into Codex

Master implementation prompt

Replace the highlighted square-bracket fields, attach the reference media, and send the completed prompt to Codex from the target repository.

Build a production-quality, scroll-controlled video component in my existing website.

GOAL

Create a cinematic section where normal page scrolling scrubs a video forward and backward. The section must pin to the viewport while active, feel smooth with a mouse wheel and trackpad, remain responsive, and integrate naturally with the existing site.

PROJECT CONTEXT

- Repository/workspace: [PROJECT PATH OR CURRENT WORKSPACE]
- Target page: [PAGE OR ROUTE]
- Placement: [WHERE THE COMPONENT SHOULD APPEAR]
- Reference video or screenshot: [ATTACH OR PROVIDE FILE PATH]
- Source video: [ATTACH OR PROVIDE FILE PATH]
- Framework: [REACT / NEXT.JS / VUE / PLAIN HTML / AUTO-DETECT]
- Required browser support: [CHROME / SAFARI / FIREFOX / MOBILE]
- Visual direction: [MINIMAL / EDITORIAL / DARK / LIGHT / MATCH EXISTING SITE]
- Visible overlay content: [NONE / DESCRIBE REQUIRED CONTENT]

WORKING METHOD

1. Inspect the repository, its AGENTS.md instructions, package scripts, current page structure, design system, responsive styles, and existing media before editing.
2. Use all relevant skills available in this Codex session, but do not invoke unrelated skills merely to increase tool usage. Read each selected skill's SKILL.md completely before using it.
3. Use the browser-control skill for visual and interaction testing. Use image/video generation or enhancement skills only when the task genuinely requires new visual assets.
4. Reuse the project's current framework, components, styling conventions, fonts, breakpoints, and dependencies. Do not introduce a new production dependency unless it is genuinely necessary.
5. Preserve unrelated behavior and user changes. Make the smallest cohesive implementation that fully solves the task.

INTERACTION REQUIREMENTS

- Use normal document scrolling-do not trap the wheel, create a nested scroll area, or prevent default scrolling.
- Pin the visual with a sticky, viewport-height stage and provide enough document height for precise frame control.
- Map the section's scroll position to normalized progress from 0 to 1.
- Scrolling down must move the video forward; scrolling up must reverse it.
- Smooth the rendered playhead with frame-rate-independent interpolation based on elapsed time.
- Coalesce video seek requests. Do not queue a new seek while the browser is still seeking; resume toward the newest target after the current seek completes.
- Match seeking frequency to the source frame rate and avoid redundant `currentTime` writes.
- Pause automatic playback. Scrolling alone must control the animation.
- Recalculate measurements after resize, orientation changes, and layout shifts.
- Remove all listeners, observers, and animation frames during component cleanup.
- Respect `prefers-reduced-motion` by removing interpolation while keeping the content usable.

VISUAL REQUIREMENTS

- The animation must be edge-to-edge with no visible border, outline, shadow, progress line, label, caption, or colored letterboxing unless explicitly requested above.
- Match the component background to the video's canvas so no seam is visible.
- **Prefer object-fit:** cover for a full-bleed result. Adjust object-position per breakpoint to keep the subject visible.
- If full subject visibility is more important than full bleed, use `object-fit: contain` and match the surrounding background exactly.
- Prevent inline-video gaps and rendering seams with `display: block` and explicit border/outline/shadow resets.
- Keep the website's global header and navigation readable without placing extra copy over the animation.
- Avoid excessive filters. If clarity needs improvement, use subtle denoising, contrast correction, high-quality scaling, and restrained sharpening.

MEDIA QUALITY AND PERFORMANCE

- Inspect and report the source video's true resolution, frame rate, duration, codec, and file size.
- Never describe an upscale as native 8K. If the source is below the requested resolution, state that clearly.
- When suitable assets are available, provide adaptive sources:
 - 7680x4320 HEVC for compatible high-resolution desktop browsers.
 - 3840x2160 H.264 as the broadly compatible desktop fallback.
 - A lightweight source for mobile and constrained devices.
- Encode scroll-scrub video with frequent keyframes so random seeking is responsive.
- Preserve aspect ratio and avoid stretching.
- Use preload and source selection carefully so mobile users do not download an unnecessary 8K asset.

- Keep React state updates out of the animation loop. Animate video time and purely visual progress with refs.
- Use passive scroll listeners, requestAnimationFrame, ResizeObserver, and GPU-friendly CSS where appropriate.
- Avoid memory leaks, render loops, layout thrashing, and continuously reading getBoundingClientRect during scroll.

ACCESSIBILITY

- Treat the video as decorative when it communicates no essential information.
- Provide a concise accessible description for the section without adding visible wording when overlays are disabled.
- Keep keyboard and touch scrolling functional.
- Do not rely on motion alone for essential navigation or actions.

RESPONSIVE BEHAVIOR

- Test at desktop and mobile sizes.
- Keep the visual full-bleed without accidental borders or differently colored side bands.
- Ensure the main subject remains visible at every breakpoint.
- Use the high-resolution source only where the viewport and codec support justify it.

VERIFICATION

Before finishing:

1. Run the project's build, lint, and relevant tests.
2. Open the local site in an available browser.
3. Test mouse-wheel scrolling through the full sequence.
4. Confirm that video time increases smoothly while scrolling down and decreases while scrolling up.
5. Test several small wheel movements, not only one large jump.
6. Verify desktop and mobile layouts visually.
7. Confirm the expected media source and decoded dimensions in each layout.
8. Check browser console errors and warnings.
9. Confirm there is no visible text, border, line, shadow, letterboxing, or background seam when those are disallowed.
10. Iterate until these checks pass.

DONE WHEN

- The component is integrated at the requested location.
- Scrolling controls the animation smoothly in both directions.
- The correct adaptive-quality source loads.
- The visual is clear, responsive, and seamless with the page.
- No unwanted wording or decoration appears.
- Build/tests pass and browser verification shows no relevant errors.

DELIVERY

Implement the change rather than only describing it. At the end, briefly list the files changed, media variants created, verification performed, and any unavoidable source-quality limitation.

Short copy-paste version

Use this version when the repository already has clear conventions and the desired placement is known.

Add a production-quality scroll-scrubbed video section to [PAGE] at [LOCATION], using [VIDEO PATH] and matching the existing design system. Inspect the repo and AGENTS.md first, then use all relevant available skills-especially browser testing-but skip unrelated skills.

The section must use normal page scrolling, stay sticky while active, scrub forward and backward, use frame-rate-independent smoothing, coalesce seek requests, avoid layout thrashing, clean up all listeners, support reduced motion, and work with mouse, trackpad, touch, and keyboard scrolling.

Make it edge-to-edge with no visible copy, border, outline, shadow, progress line, or background seam. Preserve aspect ratio and keep the subject well composed at desktop and mobile breakpoints.

Inspect the source resolution before claiming quality. Prefer adaptive 8K HEVC, 4K H.264, and mobile sources when available; never call an upscale native 8K. Optimize for fast seeking with frequent keyframes and avoid forcing large downloads on mobile.

Run the build/tests, verify the component in a real browser at desktop and mobile sizes, test small wheel movements in both directions, confirm the selected media resolution, check console errors, and iterate until all checks pass. Preserve unrelated code and do not add dependencies unless necessary.